

Advanced Digital Design Lab
130326
Hard Coded Time Base Generator (Assignment 1)
Date: Tuesday, March 18, 2025

Submission:	By not more than 2 people together
Deadline: Preparation work	There is no official preparation work for this lab. Please read through this whole paper and get familiar with the details and with what needs to be done...
Deadline: Final report	Tuesday, April 1, 2025 (18:10) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

A. Introduction: Time-Base Generator

A **time-base generator** is a mechanism that creates a single pulse, **one clock cycle wide**, at regular pre-defined time periods (intervals). These generated pulses are used to enable counters or other synchronous mechanisms, periodically.

An example for the need of a time-base mechanism can be a time measurement system that has to count seconds, minutes and hours but, its only clock source comes from a 10MHz crystal. In order to advance the seconds counter, a pulse must be created from the 10MHz input clock, once every 10,000,000 cycles. The pulse width has to be only one cycle of the 10MHz clock.

Time-base generators can be either **hard coded** or **programmable**. This lab will exercise the hard coded time base generator.

B. Hard Coded Time-Base Generator

The hard coded time-base generator is a counter that generates a TC (Terminal Count) pulse after a fixed number of clock cycles. If the frequency of the input clock is changed, then the time-base generator is no longer accurate to how it was originally designed.

Block diagram of a Hard Coded Time-Base Generator is presented in the following diagram:

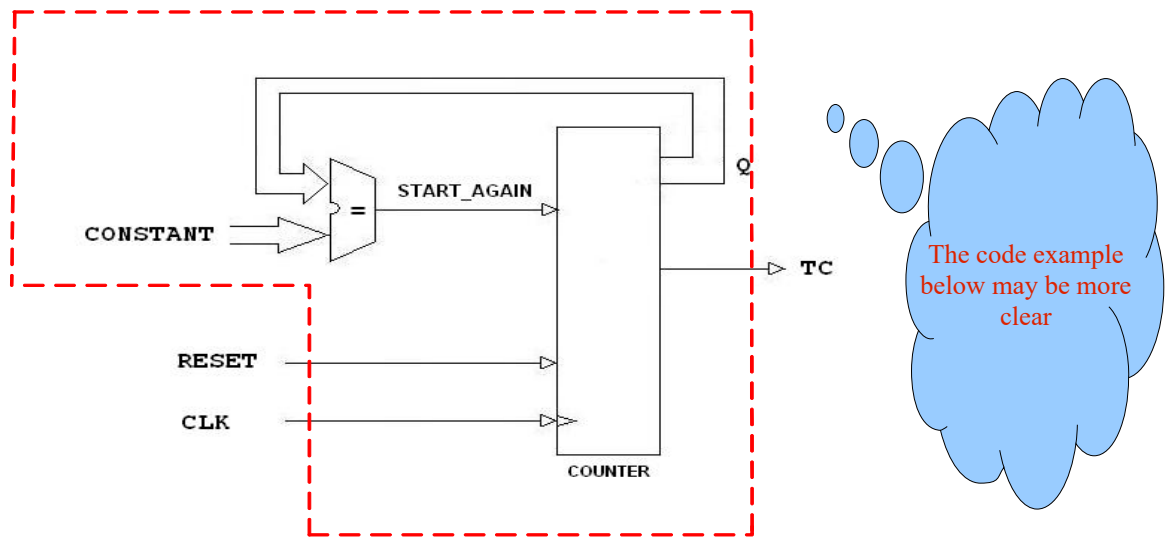


Figure 1: Hard Coded Time-Base Generator

The constant is internal (hard coded in the design source). The Q signal will be left as an internal signal. The entity shall therefore contain only three signals: CLK, RESET and TC.

Generally speaking, the hard coded time base generator is a counter. However, the counter's step (Q) is treated as an internal signal (i.e., it is usually not reflected as an output of that counter's entity). Clock, Reset and TC are the only I/Os usually required for that time base generator.

The VHDL process for the hard coded time-base generator that counts **10000** (for the example) has to be written as follows:

```
process
begin
  wait until rising_edge(CLK);
  if (RESET = '1') then
    Q <= (others => '0');
  else
    if (Q < 9999) then
      Q <= Q + 1;
    else
      Q <= (others => '0');
    end if;
  end if;
end process;

TC <= '1' when (Q = 0) else '0';
```

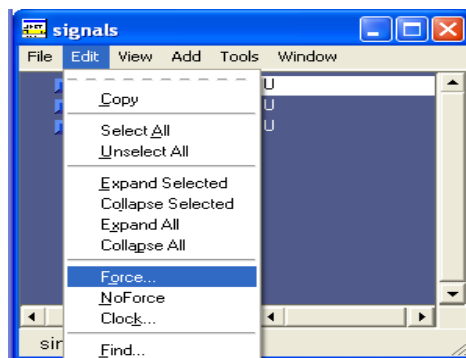
C. Design and Simulation Requirements for the Lab

1. Write the VHDL source code for a hard coded time-base generator based on your previous knowledge of writing counter's description:
 - Active high synchronous reset;
 - The TC (Terminal Count) output signal must be generated every **100 cycles** of the input clock. (This can then act as a clock-enable input to another design/system/entity.)
 - Make sure that the "Q" signal is internal (and does not show in the entity)!
2. Prepare an appropriate testbench accordingly

D. Simulation

- Simulate the design to show a number of TC pulses
- Restart the simulation and force the (internal) counter in such that after a few cycles of counting you force a value close to the final counter value. Then let the counter to go ahead and count from the forced value (we may call it "קפיצת הדרך")

Just as a reminder, in order to force a signal in ModelSim, select the relevant signal in the wave or objects window. Click on the right mouse in order to get a menu similar to the following menu:



1. Select the "Force" option and a next window will open up to allow you to enter the value that you would like to force on the signal. You then need to run the simulation some time (say one or to clock cycles).
2. In order to cancel the force, select the relevant signal again, using the same method, and the "NoForce" option from the menu. Continue to run the simulation from that point.

E. Synthesis and Place & Route

- Synthesize the design with Precision
- **Prepare the design for the Basys3 board (using an appropriate XDC file) – the TC has to go out from one of the PMOD ports**
- Implement (place & route) the design by using Vivado (as "Post Synthesis Project")

F. Your lab report must include the following:

1. VHDL sources of your design and testbench.
2. Waveforms presenting the output pulse (TC) at the specified time (VHDL source-level simulation).
3. **XDC file for Vivado** (in your report, explain the considerations for the pin locations)
4. Screenshots of “RTL Schematics” and “Technology Schematics” from Precision
5. Screenshots of “Device View” – focused on the design” and “Pin Location”, taken from Vivado
6. A question of understanding: How many 10MHz clock cycles will it take to output a TC pulse every 1 second?